

Program : Diploma in Chemical Engineering	
Course Code : 3071	Course Title: Chemical Engineering
Semester : 3	Credits: 3
Course Category: Program Core	
Periods per week: 3 (L: 2 T: 1 P: 0)	Periods per semester: 45

Course Objectives:

- To develop an understanding of thermodynamic systems.
- To enable the students to use thermodynamics concepts in chemical engineering applications.
- To provide an overview of chemical reaction engineering.

Course Pre-requisites:

Topic	Course code	Course name	Semester
Knowledge of basic Mathematics		Mathematics I	1
		Mathematics II	2
Knowledge of basic Physics		Applied Physics I	1
		Applied Physics II	2
Knowledge of basic chemistry		Applied Chemistry I	1
		Applied Chemistry II	2

Course Outcomes :

On completion of the course student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Interpret the fundamental concepts of thermodynamics.	9	Applying
CO2	Interpret the fundamental principles of energy balance and laws of thermodynamics.	16	Applying
CO3	Summarize the applications of thermodynamics in real world systems and thermodynamic properties of fluids	12	Understanding
CO 4	Outline the fundamental concepts of reaction engineering.	8	Understanding

CO – PO Mapping

Course Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1	3						
CO2	3						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcome	Description	Duration (Hours)	Cognitive Level
CO1	Interpret the fundamental concepts of thermodynamics.		
M1.01	Summarize the scope of thermodynamics.	2	Understanding
M1.02	Summarize the properties of a thermodynamic system.	1	Understanding
M1.03	Analyse state of a system and phase rule	1	Understanding
M1.04	Interpret Zeroth law of thermodynamics	1	Understanding
M1.05	Interpret first law of thermodynamics.	4	Applying
Contents: Scope of thermodynamics - Definitions and fundamental concepts in thermodynamics - System – Process - Surroundings - Homogeneous and Heterogeneous systems – Closed, Open, Isolated and Adiabatic systems. Thermodynamic state and properties - Intensive and Extensive properties-State and Path functions - Heat and work. Steady state – Equilibrium state - Phase rule Temperature and Zeroth law of thermodynamics – Heat reservoir – Heat engine – Heat pump - Reversible and irreversible processes. Statements of first law of thermodynamics – Kinetic energy – Potential energy - Internal energy – Cyclic process- Non flow process - Enthalpy - Simple numerical - Flow process – Heat capacity .			

CO2	Interpret the fundamental principles of energy balance and laws of thermodynamics.		
M2.01	Outline P-V-T behavior of pure fluids	2	Understanding
M2.02	Summarize thermodynamic processes.	1	Understanding
M2.03	Interpret enthalpy changes accompanying chemical reactions.	2	Understanding
M2.04	Interpret Hess's law of constant heat summation.	4	Applying
M2.05	Identify the limitations of first law of thermodynamics.	1	Understanding
M2.06	Interpret second law of thermodynamics.	4	Applying
M2.07	Interpret third law of thermodynamics.	1	Understanding
	Series Test – I	1	
Contents: P–V–T behavior of pure fluids. Processes - Isothermal process - Adiabatic process - Isobaric process - Isochoric process. State Enthalpy changes accompanying chemical reactions: Heat of reaction - Standard heat of reaction - Heat of Formation - Standard heat of formation - Heat of combustion. Hess's law of constant heat summation - Standard heat of reaction from heats of formation. Solve problems on Hess's law of constant heat summation and Standard heat of reaction from heats of formation. Limitations of first law of thermodynamics Second law of thermodynamics - Definition of entropy – Carnot principle – Efficiency of carnot engine – Simple numerical. Third law of thermodynamics.			
CO3	Summarize the applications of thermodynamics in real world systems and thermodynamic properties of fluids.		
M3.01	Summarize the applications of first law of thermodynamics	2	Understanding
M3.02	Outline refrigeration cycles and systems	3	Understanding
M3.03	Outline air standard cycles.	4	Understanding
M3.04	Summarize thermodynamic properties of fluids	3	Understanding
Contents: Applications of the laws of thermodynamics –Principle of fluid flow - Nozzles – Mach number – Throttling.			

Refrigeration – COP - Carnot cycle - Vapour compression cycle - Criterion for selection of refrigerants			
Steam power plant – Rankine cycle – Internal Combustion engines – Otto cycle - Diesel cycle.			
Thermodynamic properties of fluids – reference properties, energy properties, derived properties – Definition of Helmholtz free energy - Gibbs free energy – Definition of Fugacity – Chemical potential.			
CO4	Outline the fundamental concepts of reaction engineering.		
M4.01	Outline chemical reaction engineering	1	Understanding
M4.02	Summarize the concept of rate of a reaction.	3	Understanding
M4.03	Outline the type of reactors.	2	Understanding
M4.04	Outline space time and space velocity	1	Understanding
	Series Test – II	1	
Contents: Overview of chemical reaction engineering Definition of reaction rate - variables affecting rate of reaction – Speed of chemical reaction – Rate equation – State Single and multiple reactions with examples – State elementary and non elementary reactions with examples – State molecularity and order of reaction – State Rate constant & Conversion – State Zero order reaction & First order reaction. Reactors – Principle, diagram and applications - Batch reactor – PFR – MFR Space time – Space velocity.			

Text / Reference

T/R	Book Title/Author
T1	Chemical Engineering thermodynamic; K V Narayan.
R1	Chemical reaction engineering; Octave Levenspiel.
R2	Introduction to Engineering Thermodynamics; J.M. Smith, Hendrick Van Ness, Michael M. Abbott.
R3	Chemical Engineering Thermodynamics; S. Sundaram.
R4	Chemical Reaction Engineering; K. A. Gavhane

Online Resources

Sl.No	Website Link
1	https://nptel.ac.in/courses/103/101/103101004/